ONLINE APPENDIX B

No Data Left Behind – A Primer on Bayesian State-Space Models for Acceptance Sampling

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1 Model Validation

Following the Principled Bayesian Workflow (Gelman et al., 2020; Betancourt, 2018), we employed a comprehensive model validation process to validate the developed state-space model for acceptance sampling. The validation process consists of prior predictive checks, simulation-based calibration to check computational faithfulness, model sensitivity, and parameter recovery. Due to the variety of conceivable scenarios and data patterns across production contexts, we did not employ posterior predictive checks. We show the results for all parameters, however just for a subset of lots for θ.

1.1 Prior Predictive Checks

We use 250 samples from the prior distributions throughout the validation process. To verify the sampling process from the priors, we plotted the density of the samples against the theoretical distributions, as shown in Figure 6. We can see that the sampled values mostly align with their underlying theoretical distribution, thus we can assume that sampling from the priors works as desired.

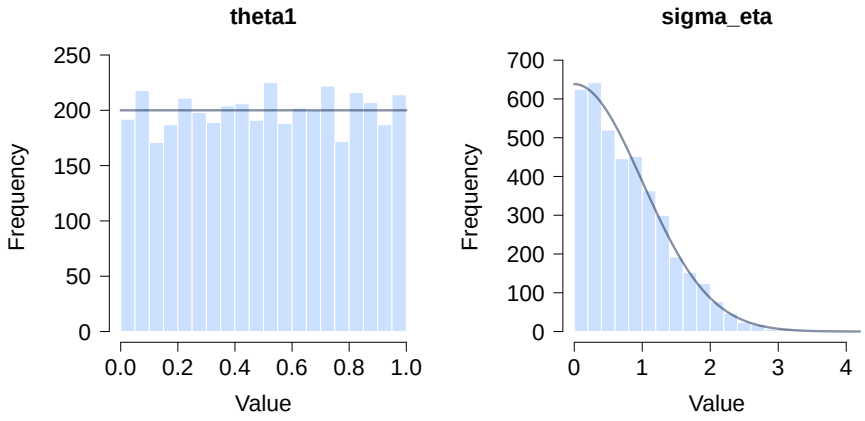


Figure 1: Samples from the prior distributions (in bins) with theoretical underlying distributions (gray density line).

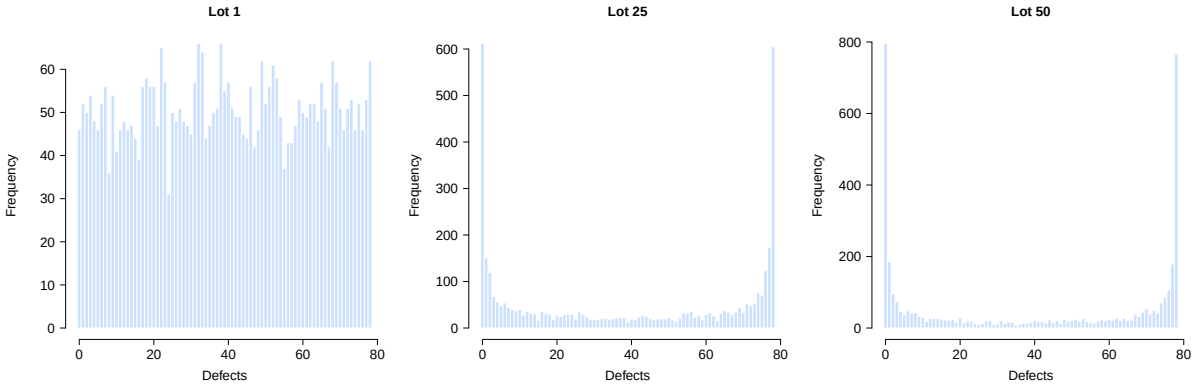


Figure 2: Prior predictive checks for lots 1,25, and 50.

As a second step, we generated data sets for each of the drawn samples from the priors. These

data sets are solely predicted by the model and its priors, without any external data introduced. This helps us to assess whether the chosen prior distributions are plausible and do not lead to unrealistic data patterns. From the chosen priors, we expect uniform distributions across many data sets, as the initial θ is assigned a uniform prior and the other parameters should allow the model to walk in any direction. Figure 7 illustrates the data patterns for lots 1, 25 and 50. Lot 1 shows the expected uniform shape. However, the parameter θ becomes increasingly U-shaped at later lots in the time series. This is an artifact due to θ operating on the logit space, and should not affect any estimates after data is introduced.

1.2 Simulation-based calibration

The introduced state-space model is estimated using a Markov Chain Monte Carlo (MCMC) algorithm. Simulation-based calibration allows us to test wether the implementation yield correct posterior approximations. This is crucial to check before applying the model to real data. Talts et al. (2018) introduced *Simulation-based calibration* (SBC), a procedure to identify sources of poorly implemented analyses including biased computational algorithms or incorrect model specifications. In SBC, artificial data sets are simulated based on the prior distributions; we can use the ones again we simulated for the prior predictive checks. The model is fitted to the simulated data. This process is repeated multiple times and results in an ensemble of posteriors distributions, one for each of the data sets (250 in this case). As no external information is added and all posterior distributions are based on draws from the priors, the resulting posterior distributions should resemble the prior distributions. Following Talts et al. (2018), we take the simulated prior parameter values and test where they are located within the estimated posterior distribution for every simulated data set; which is executed by calculating how often the prior sample is smaller than the posterior estimate. This results in a rank statistic for each parameter of the model. If the sampling works as desired and the posterior resembles the prior, the resulting overall rank statistic is approximately uniformly distributed. Consider Figure 8, we can see that all rank statistics show the desired uniform pattern, with only small deviations likely due to sampling variation.

1.3 Model Sensitivity and Parameter Recovery

When investigating model sensitivity and parameter recovery, we test whether the prior distributions are effectively updated given the expected data and the model. We examine posterior z-scores (match of estimated posterior mean and the true parameter value) and posterior contraction estimates (reduction in uncertainty from prior to the posterior). With these we are able to identify potential issues such as an overly complex model structure (Schad et al., 2021). Further, we visualize the true vs. estimated values to assess parameter recovery in an intuitive way.

Figures 9 and 10 show the outcome. We can see that the outcome of the model sensitivity shows the desired patterns across the parameters, with the strongest pattern for the θ parameters. The other parameters are more diffuse, but still in the desired area. Further, the parameter recovery plot shows that all correlations are sufficient, ranging all above 0.7.

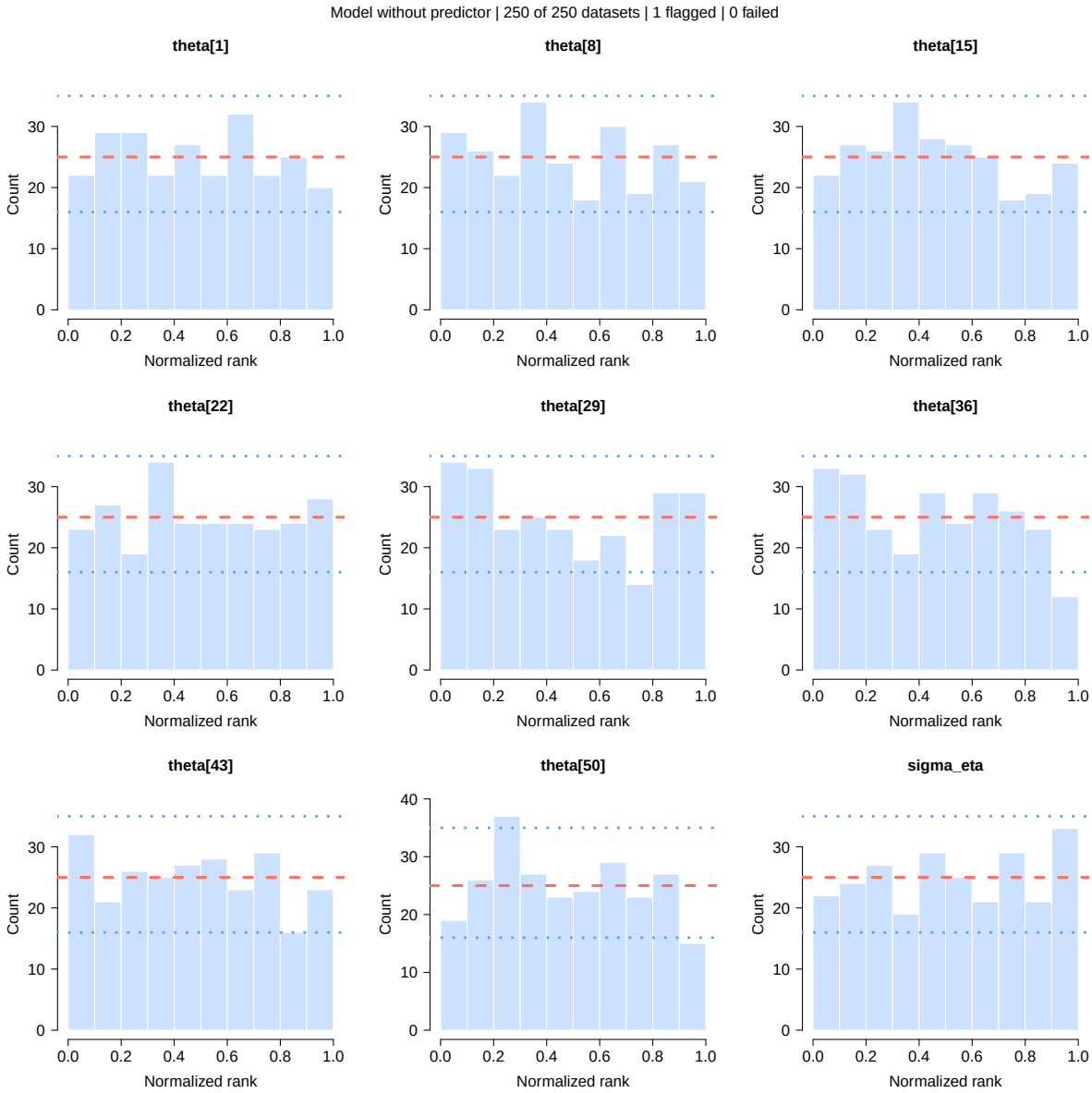


Figure 3: Simulation-based calibration outcome. We can see that all rank statistics show the desired uniform pattern, with only small deviations likely due to sampling variation.

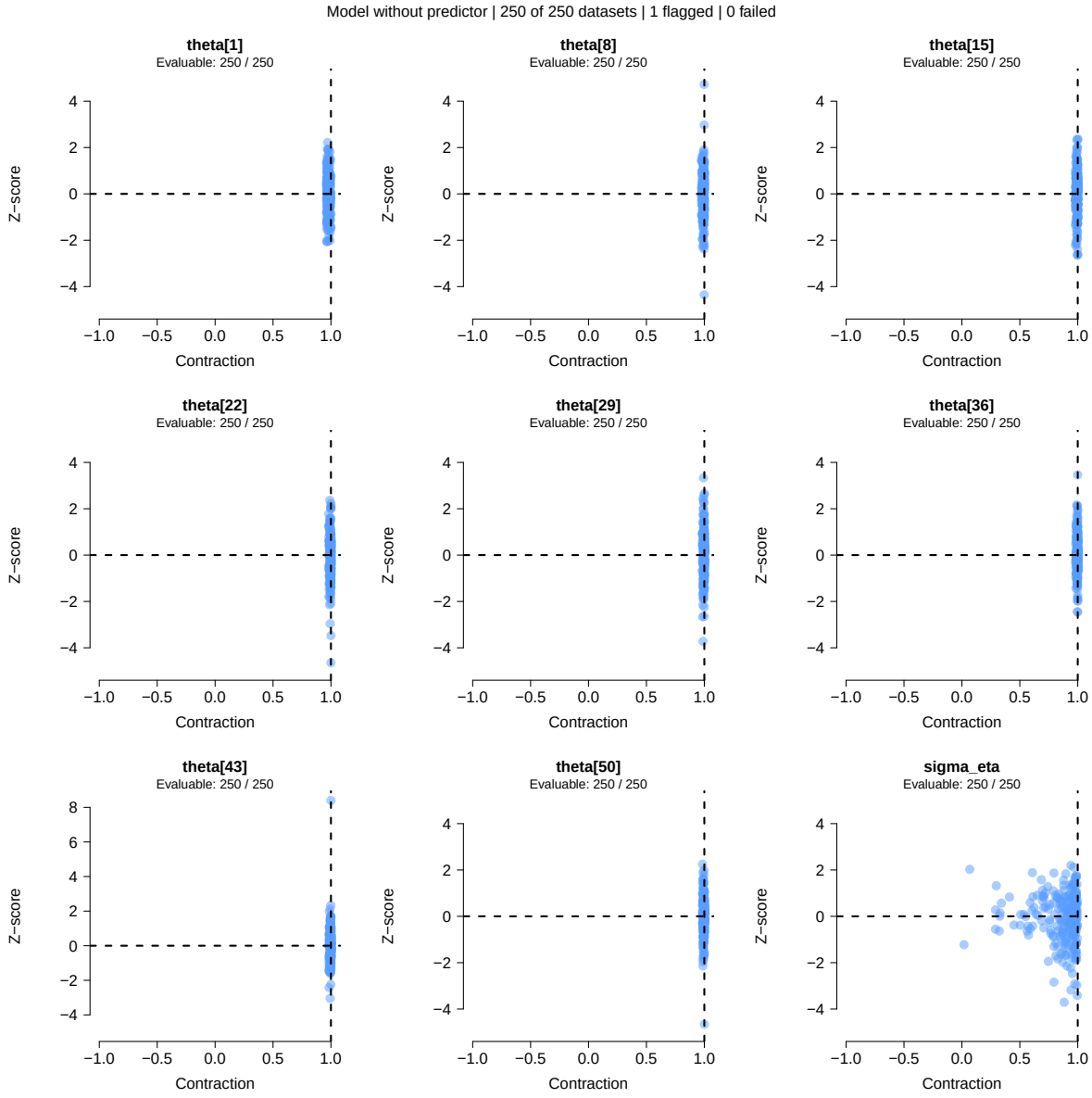


Figure 4: Scatter plot of posterior z-scores and contraction estimates.

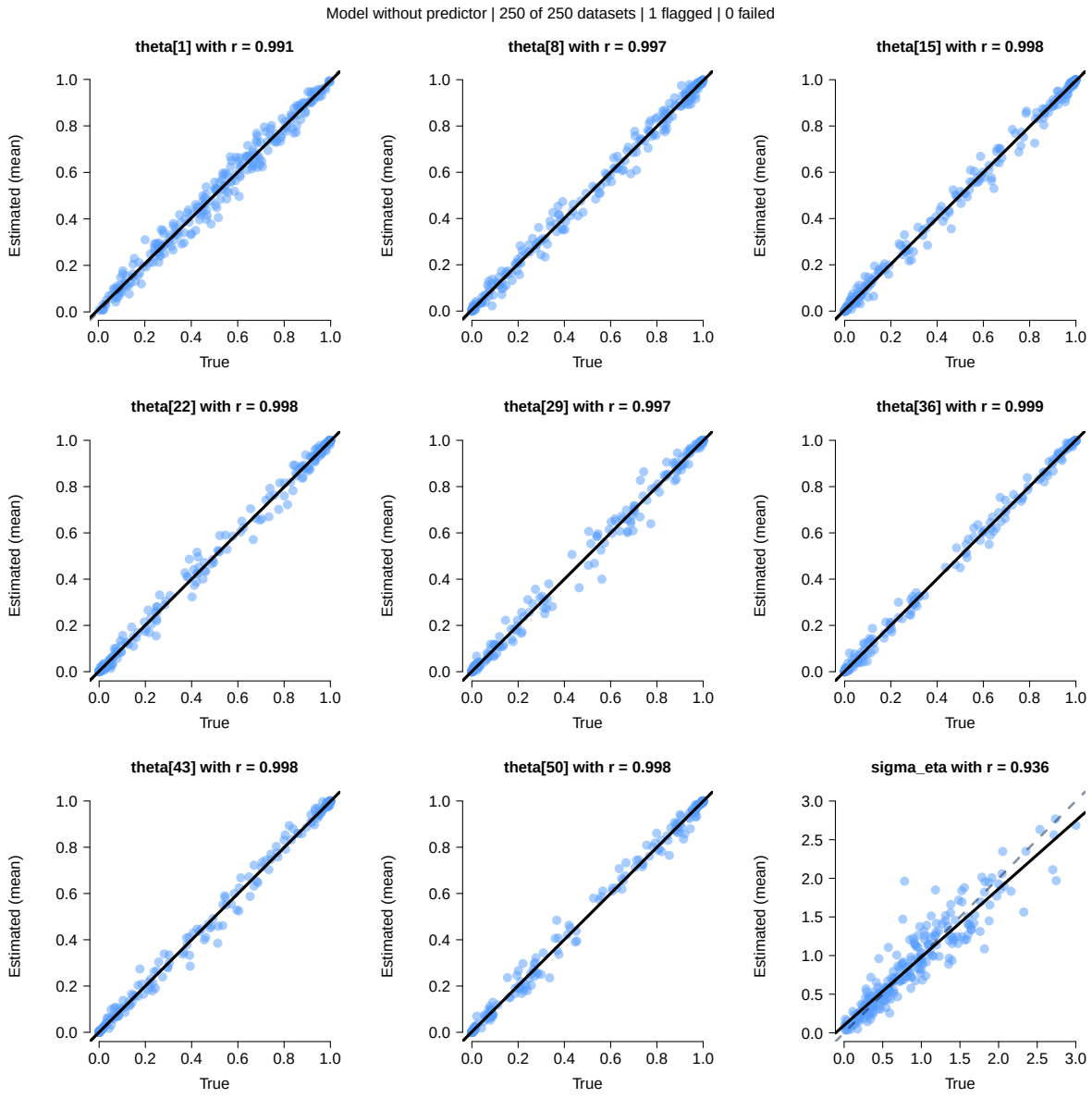


Figure 5: Correlation between true and estimated values. Line illustrates correlation.

Predictor Model plots

The plots for the predictor model look satisfactory, too. The only deviation is the first lot in the prior predictive checks. However, this U-shape already in the initial plot is introduced by the predictor variable (with a wide prior) already affecting the initial defect rate.

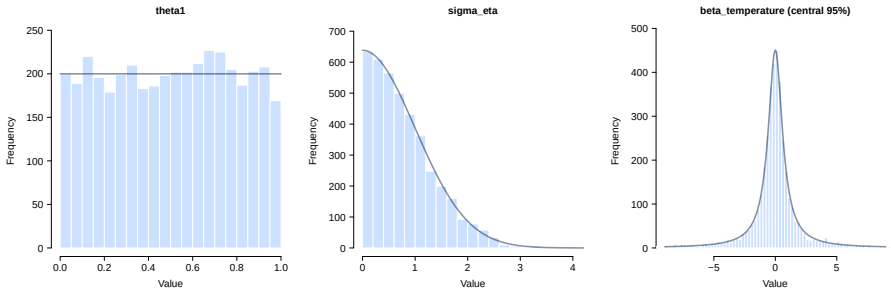


Figure 6: Samples from the prior distributions (in bins) with theoretical underlying distributions (gray density line).

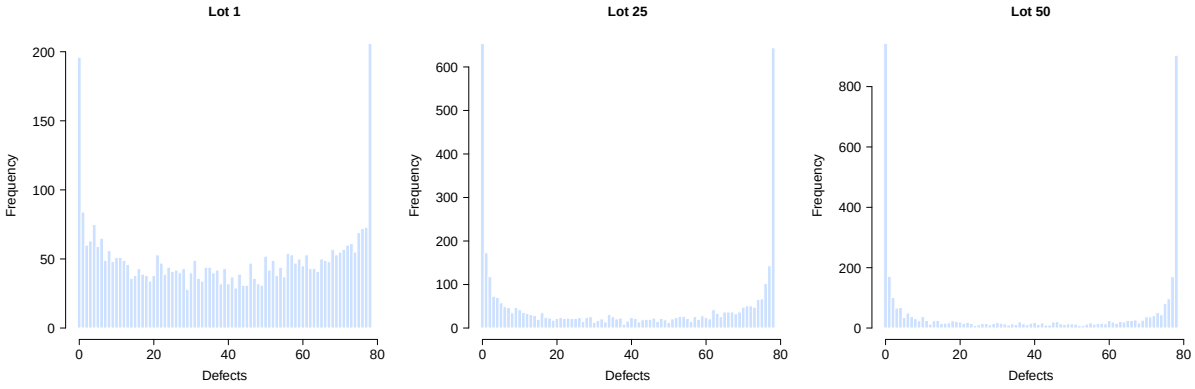


Figure 7: Prior predictive checks for lots 1,25, and 50.

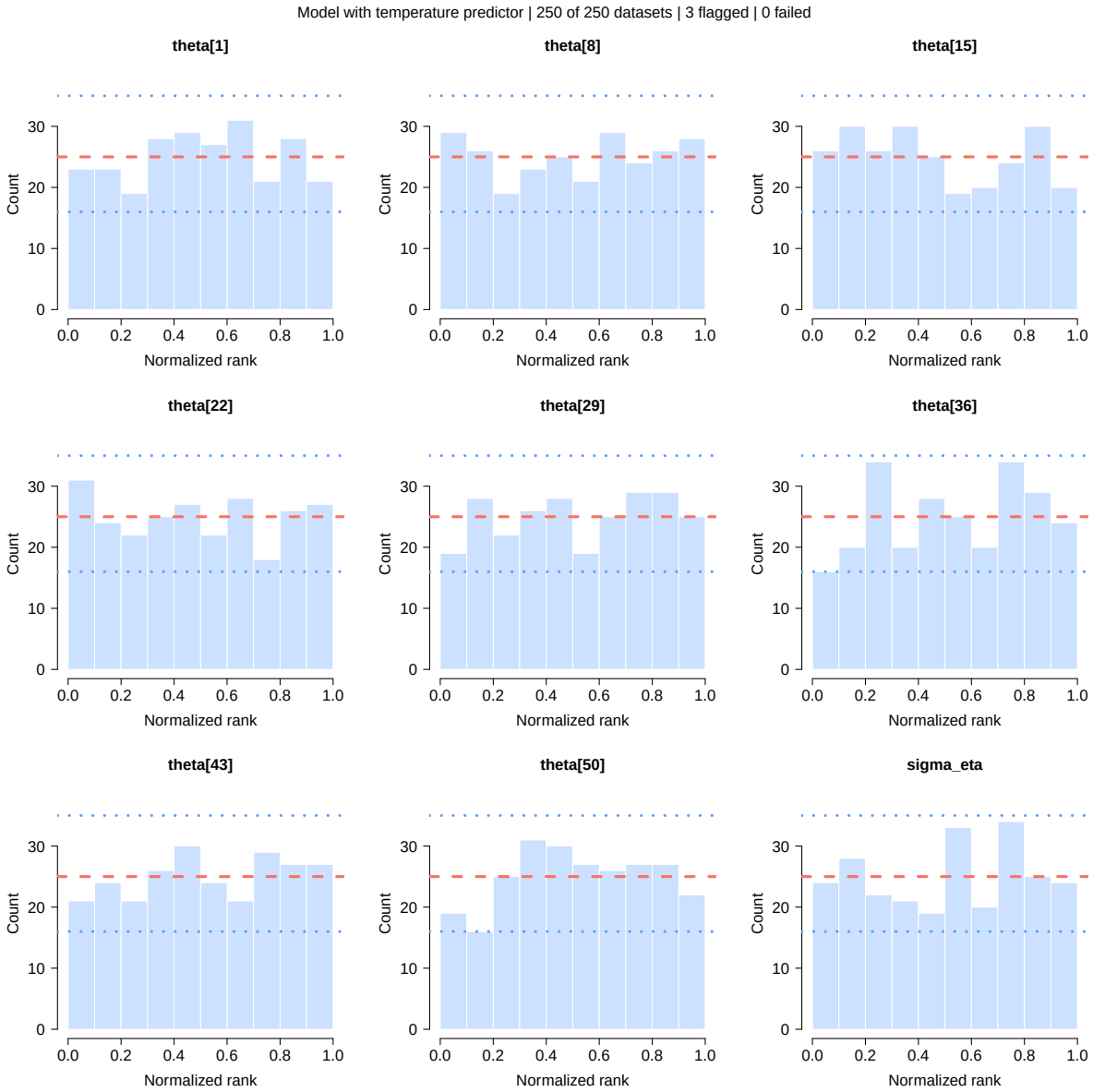


Figure 8: Simulation-based calibration outcome. We can see that all rank statistics show the desired uniform pattern, with only small deviations likely due to sampling variation.

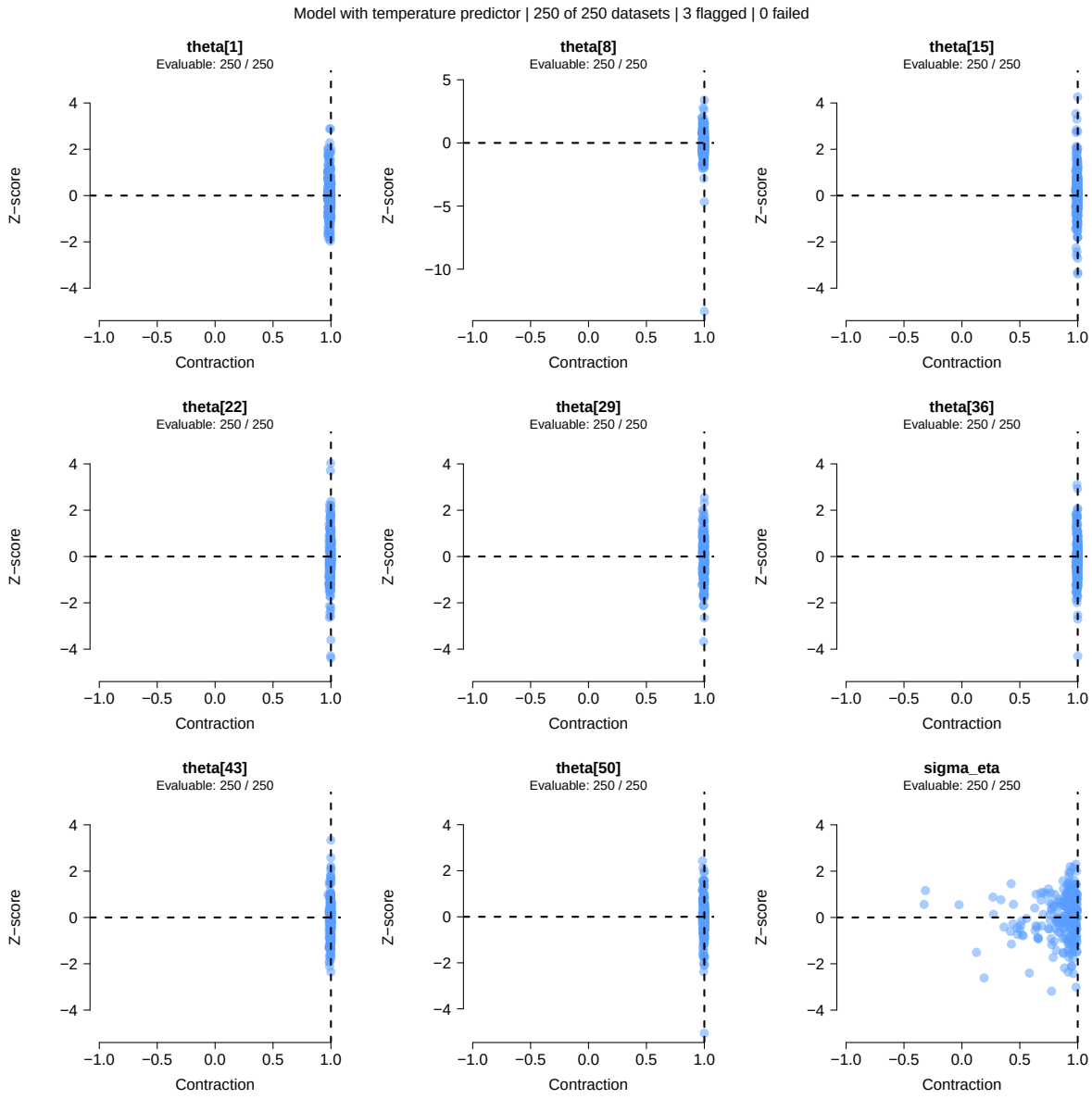


Figure 9: Scatter plot of posterior z-scores and contraction estimates.

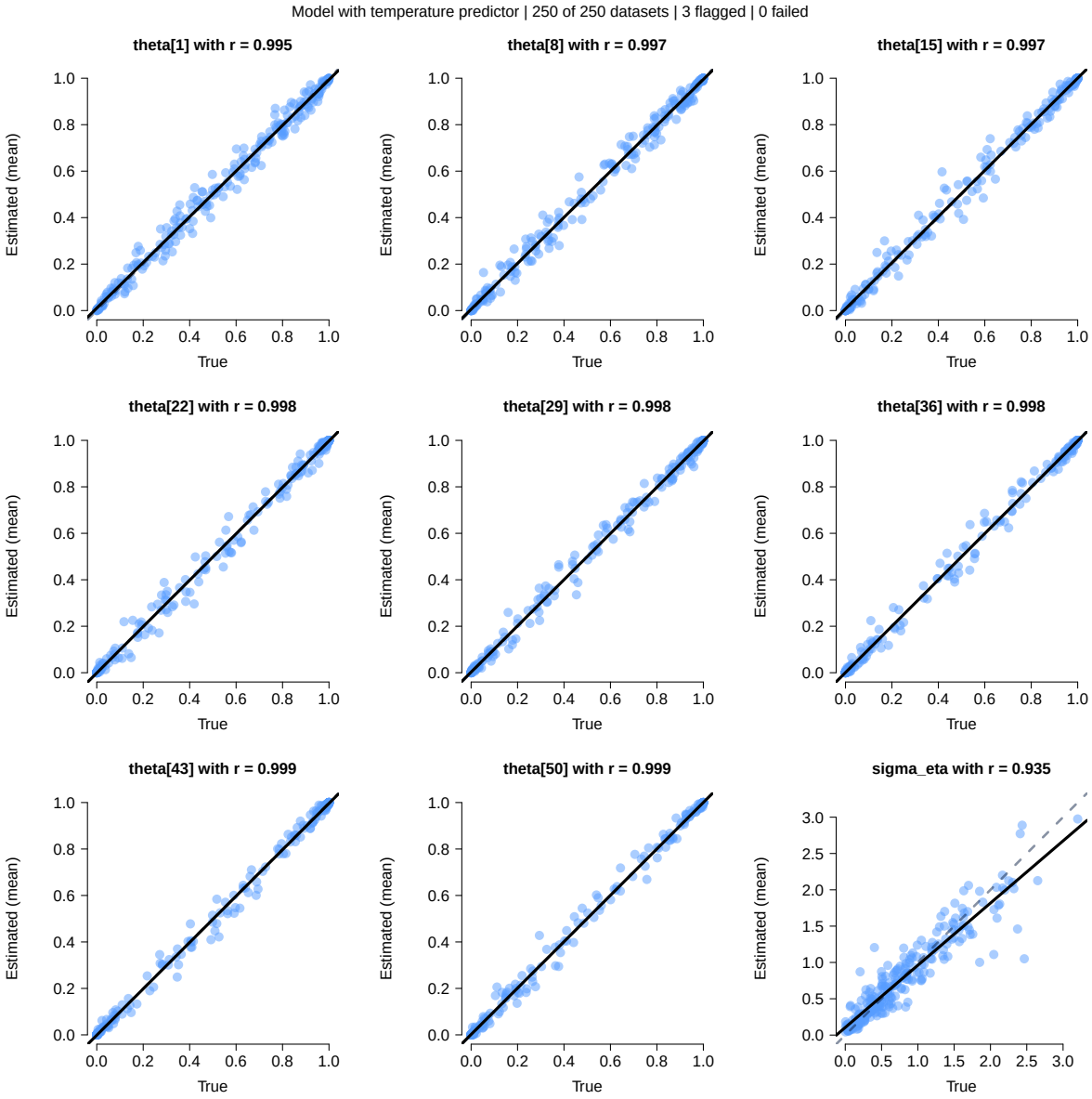


Figure 10: Correlation between true and estimated values. Line illustrates correlation.

References

Betancourt, M. (2018). Towards a principled Bayesian workflow (pystan).

Gelman, A., Vehtari, A., Simpson, D., Margossian, C. C., Carpenter, B., Yao, Y., Kennedy, L., Gabry, J., Bürkner, P.-C., and Modrák, M. (2020). Bayesian workflow. *arXiv preprint arXiv:2011.01808*.

Schad, D. J., Betancourt, M., and Vasishth, S. (2021). Toward a principled Bayesian workflow in cognitive science. *Psychological Methods*, 26(1):103–126.

Talts, S., Betancourt, M., Simpson, D., Vehtari, A., and Gelman, A. (2018). Validating Bayesian inference algorithms with simulation-based calibration. *arXiv preprint arXiv:1804.06788*.